Building Denmark's power system from scratch

Investment

Reading. *Energy System Models* (ESM below), section 7.

The Lab and the Explorer. This worksheet holds the questions of lecture 6 in the Model Lab, where you answer them a part at a time. The figures they need are in the Model Explorer: every scenario of the section 7 model, solved in advance and drawn figure by figure, with a description of the model under "About this model".

The Model Lab	*championape-model-lab.share.connect.posit.cloud*
The Model Explorer	*championape.github.io/EnergyEconomics-explorer*

Each question names the view it needs: the Explorer's tab, the figure, and the scenario where it is not the reference case. A prediction asks whether each quantity rises, falls or stays about the same against the reference case. Make it before you look, then check it in the Explorer, where every scenario can be compared with the reference.

Part 1 · ESM §7.2–7.3 30 min

Read the reference system

This part reads the reference case the way a modeller reads any solved capacity-expansion model: what is built and why it stops there, what the investment condition says about each technology, and what sets the price, hour by hour. Derive the condition first, then read it in the model. Keep the Explorer open beside the worksheet: several questions ask you to find your own evidence there. The last question is a stretch, for those who finish early.

1. *Optimal investment.* The reference case of section 7: 85 €/t on every emitter in every zone, every zone's demand at its 2050 level, and Denmark rebuilt from scratch inside a twelve-zone neighbourhood that may add to its own fleets.

> **In the Explorer:** Overview › *What Denmark builds*, the reference case. The tick marks each technology's potential.

(i) Derive Result 6 for a generator g. Use the first-order conditions for its capacity $\bar{q}_g$ and for its generation $q_{g,h}$, and complementary slackness on $0 \le q_{g,h} \le \gamma_{g,h}\,\bar{q}_g$, to show that

$$\overline{\beta}_g - \underline{\beta}_g = \sum_h \gamma_{g,h} \max\left(\lambda_h - c_g - \tau e_g,\ 0\right) - \left(a(r, n_g) + f_g\right) I_g,$$

where λ_h is the price in g's zone. Say in words what the sum on the right-hand side is. *at most 250 words*

(ii) For which of these is $\overline{\beta}_g > 0$ in the reference case? Tick all that apply.

- (a) Onshore wind in DK1
- (b) Onshore wind in DK2
- (c) Solar in DK1
- (d) Solar in DK2

(e) Offshore wind in DK1
(f) Offshore wind in DK2
(g) Gas peakers in DK2
(h) Nuclear in DK1

(iii) When capacities are chosen optimally, how is a technology's capture price related to its levelised cost? Distinguish the three cases of Result 6, and find an example of each in the reference case, using the Explorer. *at most 150 words*

2. *The generation mix.*

(i) Across Denmark, a MWh of solar costs less to produce than a MWh of onshore wind. Yet, onshore wind is the first to use its full potential and earns the highest rent. What must be true?
(a) Onshore wind runs more hours a year than solar, which makes a MWh of it cheaper than its levelised cost says.
(b) Onshore wind's land is scarcer, and the land rent is what the turbine pays for it.
(c) Onshore wind's output sells at a higher average price than solar's, since more of it comes in hours when power is worth more; its value factor is higher.
(d) Solar is curtailed more often, so it sells less of what it could produce.

(ii) A peaker has running cost c and annual fixed cost F per MW. It is built below its potential, so Result 6 holds with $\overline{\beta} = \underline{\beta} = 0$, and it is always available ($\gamma_h = 1$). Which condition must the year's prices λ_h satisfy?
(a) $\sum_h (\lambda_h - c) = F$
(b) $\sum_h \max(\lambda_h - c, 0) = F$
(c) $\max_h \lambda_h - c = F$
(d) $\frac{1}{8760} \sum_h \lambda_h = c + F/8760$

(iii) What drives the Danish investments in gas turbines in the reference system? *at most 120 words*

(iv) Denmark is rebuilt from scratch, while its neighbours keep the fleets they have and may add to them. How does their generation mix differ from Denmark's, and how does one of those differences shape what Denmark builds or how it trades? Find your evidence in the Explorer. *at most 120 words*

3. *What sets the price.*

In the Explorer: Operation › *What is at the margin, month by month*, zone DK2: what is at the margin of the DK2 price, month by month, in the reference case.

(i) Which statements about the Danish price in the reference case hold? Tick all that apply.
(a) A Danish gas plant is at the margin in about a tenth of the hours.
(b) In about seven hours in ten the price is set together with a neighbour, through an interconnector carrying power below its capacity.
(c) In the winter months a Danish gas plant is at the margin in most hours.
(d) Wind and solar in surplus push the price to near zero in every month of the year, most often in May and June.

(ii) A Danish gas plant or Danish flexible demand is at the margin of the DK2 price in only about 150 hours of the year; in most other hours trade with a neighbour sets the price. Why do Danish plants and consumers set the Danish price so rarely?
(a) The model lets Danish plants run only when the neighbours cannot supply.
(b) Prices are set by the average cost of the plants running, and Danish plants are a small share of them.

(c) The Danish gas plant costs more to run than the neighbours' plants, so it never runs.

(d) Denmark is small beside the interconnectors around it. As long as they are not full, the price is set in the larger market they join, by plants abroad; Danish resources set it only when the lines are full.

(iii) In the hours with the highest prices, the DK2 price reaches about 2,000 €/MWh, far above the running cost of anything in the system. What is at the margin in those hours, and what decides how high the price goes? Find your evidence in the Explorer.

at most 120 words

If you finish early

Optimal battery investment. In section 4's notation, a battery with power rating $\bar{s}$ (MW) and duration $\bar{\delta}$ (hours) charges g^c_h, discharges g^d_h and holds soc_h in store, at an operating cost c^s per MWh charged or discharged, subject to

$$soc_h = soc_{h-1} + \eta_c\, g^c_h - g^d_h/\eta_d \quad (\theta_h),$$

$$0 \le g^c_h \le \bar{s} \quad (\underline{\phi}^c_h, \overline{\phi}^c_h), \qquad 0 \le g^d_h \le \bar{s} \quad (\underline{\phi}^d_h, \overline{\phi}^d_h), \qquad 0 \le soc_h \le \bar{\delta}\,\bar{s} \quad (\underline{\psi}_h, \overline{\psi}_h).$$

θ_h is the value of a MWh in store at the end of hour h. In section 7, $\bar{s}$ becomes a variable: a MW of battery costs F_s a year, the cells for its $\bar{\delta}$ hours included, and $\bar{s} \ge 0$ has the multiplier $\underline{\beta}_s$.

(i) The first-order conditions for charging, discharging, the state of charge and the power rating are

$$g^c_h: \; -c^s - \lambda_h + \eta_c\,\theta_h + \underline{\phi}^c_h - \overline{\phi}^c_h = 0,$$

$$g^d_h: \; -c^s + \lambda_h - \theta_h/\eta_d + \underline{\phi}^d_h - \overline{\phi}^d_h = 0,$$

$$soc_h: \; -\theta_h + \theta_{h+1} + \underline{\psi}_h - \overline{\psi}_h = 0,$$

$$\bar{s}: \; -F_s + \sum_h \left(\overline{\phi}^c_h + \overline{\phi}^d_h\right) + \bar{\delta} \sum_h \overline{\psi}_h + \underline{\beta}_s = 0.$$

Use them and complementary slackness to show how $\overline{\phi}^c_h$, $\overline{\phi}^d_h$ and $\overline{\psi}_h$ relate to the hourly prices λ_h. *at most 250 words*

Hint: take each limit in turn, as for a generator in Result 6: the battery at the limit, at zero, or in between.

(ii) Read the condition for $\bar{s}$ for a battery that is built. What decides how much another MW of it would earn in a year, and in which hours does it earn it? *at most 120 words*

(iii) The model may build batteries of 2, 4 or 8 hours, and the reference builds only 8-hour ones. Find out in the Explorer why the shorter ones are not built. *at most 120 words*

Hint: compare what a MW of each battery would earn with its annual fixed cost (Money, "Rent per MW against annual fixed cost"), and look at how long the batteries charge and discharge, and how the price moves, through the day (Operation, "What the batteries do through the day, month by month" and "The price through the day, month by month").

Shock it: carbon, fuel and capital

1. *A higher carbon price.* The carbon price on every emitter in every zone rises from the reference 85 €/t to **225 €/t**. Predict what happens in Denmark.

	rises	about the same	falls
Danish emissions	☐	☐	☐
Solar built in Denmark	☐	☐	☐
Batteries built in Denmark	☐	☐	☐
Gas peakers built in Denmark	☐	☐	☐
The average Danish price	☐	☐	☐

Then compare in the Explorer: *Carbon price 225 EUR/t* against the reference.

2. *Denmark's marginal abatement cost curve.*

> **In the Explorer:** Sweeps: *Danish emissions* against *Carbon price, every zone*. Each point is a solved scenario. The menu above the chart changes the quantity, and the legend adds the scenarios that move a second lever too.

The figure shows Danish emissions against the carbon price. Briefly explain the shape of Denmark's marginal abatement cost curve. *at most 120 words*

3. *Expensive gas.* The note's experiment with expensive gas (section 7) doubles the price of natural gas for every gas plant the model builds, in every zone, while the plants that already exist keep paying the forward price. The carbon price stays at 85 €/t. In the Explorer this scenario is **Gas at 2 x, new plants only**.

(i) Against the reference case, predict:

	rises	about the same	falls
Gas peakers built in Denmark	☐	☐	☐
Combined-cycle gas plants built in Germany	☐	☐	☐
Danish emissions	☐	☐	☐
Flexible demand left unserved in Denmark	☐	☐	☐

Then compare in the Explorer: *Gas at 2 x, new plants only* against the reference.

(ii) A combined-cycle plant burns less gas per MWh than a peaker, so a higher gas price raises its running cost less. Doubling the gas price therefore lowers the number of hours a year above which a combined-cycle plant is the cheaper of the two, from about 2,300 to about 1,550. Yet Germany stops building combined-cycle plants, and Denmark builds more peakers. Why?

(a) Wind, solar and batteries now produce energy over thousands of hours more cheaply than a new gas plant, so gas is left with standing by for a few hundred hours, where the peaker is the cheaper plant.

(b) A peaker costs less than half as much to build, and when fuel is expensive the plant that is cheapest to build becomes the cheapest to own.

(c) A combined-cycle plant cannot start and stop as quickly as a peaker, and with expensive gas the system needs plants that run only when they must.

(d) The screening curves leave out carbon, which weighs more on the combined cycle and tips the comparison towards the peaker.

(iii) Denmark's peakers now run fewer hours and burn less gas. Why does Denmark build twice as many of them?

(a) A higher gas price raises the electricity price in every hour a gas plant sets it, so a peaker earns more on every MWh it sells, and more peakers pay for themselves.

(b) The model requires a margin of firm capacity above peak demand, and the peakers fill the gap that the combined-cycle plants leave in Germany.

(c) The combined-cycle plants Germany builds in the reference also run at full output through the winter spells. Without them capacity is short in those spells more often, so a peaker earns more than its fixed cost there, and peakers are added until it no longer does.

(d) A peaker runs so few hours that its fuel cost hardly counts, so a higher gas price makes it cheaper than everything it competes with.

(iv) In the reference, gas is the last step of supply in the merit order: above the peaker there is only demand, ready to step back at a price. When gas is expensive, what takes its place? Compare **Gas at 2 x, new plants only** with the reference in the Explorer, for Denmark and for Germany. Tick all that hold.

(a) In Denmark the peaker now costs more to run than flexible demand is willing to pay, so in a tight hour flexible demand steps back before a peaker starts.

(b) In Germany, coal and lignite plants take over most of what the combined-cycle plants no longer produce.

(c) In Germany, solar takes over most of what the combined-cycle plants no longer produce, with batteries to carry it into the night.

(d) Denmark builds more offshore wind.

4. *Cheap capital.*

(i) Capital becomes cheap: the discount rate falls from 7 % to **2 %**, for every technology. Predict:

	rises	about the same	falls
Solar built in Denmark	☐	☐	☐
Batteries built in Denmark	☐	☐	☐
Gas peakers built in Denmark	☐	☐	☐
The average Danish price	☐	☐	☐
The Danish net cost	☐	☐	☐

Then compare in the Explorer: *Discount rate 2 %* against the reference.

(ii) Why does the discount rate matter more for wind and solar than for a gas plant?

at most 80 words

Margins of response

A system adapts to a shock through its **margins of response**: building something else, storing, trading, or demand response. Every scenario in this part is also in the Explorer, where you can compare it with the reference.

1. *No batteries.*

(i) No battery can be built, anywhere. Against the reference case, predict:

	rises	about the same	falls
Gas peakers built in Denmark	☐	☐	☐
Flexible demand not served in Denmark	☐	☐	☐
Solar built in Denmark	☐	☐	☐
Danish emissions	☐	☐	☐

Then compare in the Explorer: *No batteries can be built* against the reference.

(ii) Without batteries, Danish demand response is called on **less** often, not more. Explain the mechanism. *at most 100 words*

2. *Batteries and peakers.* Batteries and peakers both supply power when wind and sun fall short. In eastern Denmark (DK2) the reference builds both: about 1.4 GW of 8-hour batteries and about 0.85 GW of gas peakers ("Natural gas" in the figures). This question asks what each does, and why one of the two would not do.

(i) The reference builds only 8-hour batteries. Now only **2-hour** batteries can be built, anywhere. Against the reference case, predict:

	rises	about the same	falls
Battery power built in Denmark	☐	☐	☐
Gas peakers built in Denmark	☐	☐	☐
Solar built in Denmark	☐	☐	☐
Emissions in all twelve zones	☐	☐	☐

Then compare in the Explorer: *Only 2-hour batteries can be built* against the reference.

> **In the Explorer:** Operation › *A week of operation*, the reference case, zone DK2, for two weeks: "A summer week (week 27)" and "A winter week (week 3)", the week with the year's highest prices.

(ii) Compare the two weeks. Which statements hold? Tick all that apply.

(a) The batteries discharge more in the winter week than in the summer week.
(b) In the summer week the batteries charge in the middle of the day and discharge later, and the peaker never runs.
(c) In the winter week the peaker runs for more than a day without stopping, twice.
(d) The peaker runs in both weeks, to cover the evening peak.

(iii) Two kinds of shortfall bring high prices. A: a few hours every evening, on most days of the year. B: a spell of several days, once a winter. Which is the cheaper way to cover each, a battery or a gas peaker?

(a) The battery for both, since it costs almost nothing to run.
(b) The peaker for both, since it costs less to build per MW.

(c) The peaker for A, the battery for B.
(d) The battery for A, the peaker for B.

3. *Demand response.*

(i) Consumers no longer respond to prices: both cheap demand blocks are removed in every zone, so demand is either served or lost at 23,400 €/MWh. Against the reference case, predict:

	rises	about the same	falls
The highest Danish price of the year	☐	☐	☐
Gas peakers built in Denmark	☐	☐	☐
Batteries built in Denmark	☐	☐	☐

Then compare in the Explorer: *No demand response* against the reference.

(ii) Brown, Neumann and Riepin (2025) argue that in a system with almost no fuel costs the price is set in most hours by **demand's willingness to pay**, or by storage, rather than by a generator's running cost. Compare the reference with the scenario without demand response, in DK2. What does the model say?

(a) Demand response changes only quantities, so the prices are the same with and without it.
(b) Demand seldom sets the Danish price itself, since trade does in most hours; but taking demand response away still reshapes prices, because it changes what is built.
(c) Flexible demand sets the Danish price in hundreds of hours, and without it those hours clear at the value of lost load.
(d) Without demand response the gas peaker sets the Danish price in most hours of the year.

4. *Denmark's borders.* Two experiments on what Denmark's borders do: shut them, or let a Danish target stop at them.

(i) Denmark is cut off from its neighbours: no interconnector, a Danish system on its own (the Great Belt between DK1 and DK2 stays). Against the reference case, predict:

	rises	about the same	falls
Offshore wind built in Denmark	☐	☐	☐
Gas peakers built in Denmark	☐	☐	☐
Danish emissions	☐	☐	☐
The Danish net cost	☐	☐	☐

Then compare in the Explorer: *Autarky* against the reference.

(ii) Instead of a carbon price, Denmark caps its **own** emissions at a tenth of what they are with no carbon policy; nobody prices carbon anywhere. Against the world with no carbon price at all, predict:

	rises	about the same	falls
Danish emissions	☐	☐	☐
Emissions of all twelve zones	☐	☐	☐
Danish net exports	☐	☐	☐
Solar built in Denmark	☐	☐	☐

Then compare in the Explorer: *Danish budget at 10 %* against *Carbon price 0 EUR/t*.

(iii) At a budget of half Denmark's unpriced emissions, nine tenths of the Danish cut leaks abroad; at a twentieth, none does, and emissions abroad even fall a little. Why does leakage shrink as the budget tightens?

(a) A tighter budget raises the carbon price in the neighbouring zones.

(b) A tight budget forbids imports.

(c) A loose budget is met cheaply by importing more and running Danish peakers less; a tight one can only be met by building clean capacity at home, which also exports clean power and displaces fossil plants abroad.

(d) Leakage is a constant share of the cut; the differences are noise.